EPOCH bug fixes and incremental improvements

Fork-to-upstream status register

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Abstract

This register identifies selected bug fixes and modest numerical or usability improvements in `epoch_dev`, and asks a precise question for each: is the correction present in the official EPOCH `main` branch? It includes the laser absorption diagnostic fix merged through PR #896 and the exact nonlinear Breit–Wheeler (NBW) rate tables in the fork. It is an inventory of source changes, not a claim that all fork features have been proposed upstream.

1 How status was checked

The official `main` tip was `f294c484f76dff0777d5cc0d50b38506a2b049ff` on 30 October 2026 (`git ls-remote` against `epochpic/epoch`). The audited fork tip was `b908c85911b314f6e07ed9a31caf0396612ffd83` on `spin`. "Yes" means the upstream merge commit is an ancestor of official `main`. "No" means the relevant fork commit is not an ancestor and the named table, switch, guard, or algorithm is absent from the official `main` source at that tip. A merge into `4.20-devel` alone is *not* counted as present in `main`. Other official development branches are mentioned only where their status was separately checked.

The scope is substantive source fixes and small improvements visible in the fork history. Major new physics packages and uncommitted prototype code are outside this register. Dimensions refer to the implementation at the cited fork commit; they do not imply equivalent support in the other dimensions. These are source and history checks, not fresh EPOCH simulation results.

Table 1: Selected changes and official `main` status. "2D" means `epoch2d`; "all" means 1D, 2D and 3D.

Change	Location and correction	Evidence	Main
Laser absorption diagnostic	`epoch{1,2,3}d/src/laser.f90`: correct injected-energy sign at maximum boundaries.	`0984b1d31`; PR #896	**Yes**
Photon-probe ID deadlock	Official all-dimension collective call fix reached `4.20-devel`; fork also repairs its 2D photon path.	`241d6fb9c`; `3e4734522`	**No**
Particle ID lifetime	2D: retain the maximum ID through restart and file loading; prevent reuse.	`3e4734522`	**No**
Higuera–Cary pusher	All: use the correctly normalised magnetic rotation coefficient for interpolated fields.	`d86ceceab`	**No**
Trident guard	All: obey `produce_pairs = F` in the trident optical-depth/event path.	`1a88d4afa`	**No**
NBW photon species	2D `photons.F90`: pass the active photon species to pair generation.	`2b18ef6b4`	**No**
`identify` warning	All species deck parsers: warn when `identify` replaces explicit charge or mass.	`b908c8591`	**No**
Spin restart	2D `setup.F90`: restore spin state on restart.	`a66b394cb`	**No**
Pair-channel boosting	All: repair Breit–Wheeler boost handling, muon sampling, Bethe–Heitler scaling and additive optical depths.	`3e9f3121f`; `99ce3c685`	**No**
Exact NBW rate table	2D `TABLES/pairprod_lcfa.table`, selected by `pair_tables = lcfa`.	`848d21a4b`; `fd56a2e08`	**No**
NBW energy split	2D polarisation-resolved pair energy tables and sampling.	`dfb93b141`; `2b18ef6b4`	**No**
NLCS quantile sampling	2D spin-resolved inverse-CDF tables selected by `use_nlcs_quantile_tables`.	`6610b3768`	**No**

Change	Location and correction	Evidence	Main
No-transition update	2D sub-step spin/polarisation update around QED emission.	`944fff881`	**No**
Radiation reaction	2D full Landau–Lifshitz terms, alternative Gaunt fit and midpoint kick.	`dfa48930a`	**No**
Laser half-cell sampling	2D custom profile sampling coordinates corrected.	`1b307b37d`	**No**
Laser loading deadlock	2D MPI profile and phase-file load made collective-safe.	`5a77c367b`	**No**
Per-laser file storage	2D: distinct laser blocks retain distinct imported profiles.	`945fd060d`	**No**
3D profile re-slabbing	3D: reload the spatiotemporal window as domain slabs move.	`f572d035c`	**No**

2 Two upstream merges that require different labels

Laser absorption, already on `main`. Tom Goffrey merged PR #896 into `4.20-devel` on 10 April 2026 as `0984b1d31`. The patch corrects the sign of injected energy at `x_max`, `y_max` and `z_max`; the previous negative energy could give a zero absorption coefficient (issue #893). The merge is an ancestor of official `main` and is listed in the v4.20.1 release notes. No fork-only fix remains to upstream for this item.

Photon-probe IDs, merged to development but absent from `main`. PR #906 merged `241d6fb9c` into `4.20-devel` on 1 June 2026, after the current official `main` tip. In the fork, `3e4734522` moves the 2D photon ID collective call ahead of the empty-rank return and additionally fixes ID counter persistence across restarts and file-based particle loading. The official PR and the fork commit are related but *not equivalent in scope*; the latter's restart/file-ID changes should be reviewed separately.

3 Numerical and QED improvements in the fork

NBW tables. Official `main` reads `pairprod.table`; the 2D fork adds `pairprod_lcfa.table` and the `pair_tables = legacy | lcfa` selector. The legacy rate uses Erber's approximate analytic expression; `lcfa` selects a numerically tabulated locally constant field approximation (LCFA) rate. The fork also provides parallel/perpendicular NBW energy-split tables. With spin QED off, `legacy` keeps the old rate and `energy_split.table`, while `lcfa` selects the LCFA rate and unpolarised split. With spin QED on, `lcfa` is the default, and `legacy` restores only the old rate; the spin-resolved split remains. This changes physics results, so published runs should record the selector and table hashes. The generator scripts for these tables are currently untracked in the local source tree; the committed tables and reader are present, but clean-checkout regeneration has not been demonstrated here.

NLCS sampling and time stepping. The 2D fork adds spin-resolved NLCS quantile tables and a switch to use the previous lookup. It also adds a sub-step no-transition update. Neither is part of official `main`; these affect spin-aware QED paths and should not be described as global corrections to every EPOCH emission mode. The legacy `ksi_sokolov.table` remains relevant to spin-off runs.

Radiation reaction. The 2D fork's `use_full_landau_lifshitz`, `gaunt_factor` and `rr_midpoint` options supplement EPOCH's original reduced force and explicit kick. Defaults preserve the old branch behaviour; enabling these options changes the physical model or time integration. There is no 3D implementation of this fork change in the audited `spin` branch.

4 Correctness fixes and operational limits

The Higuera–Cary correction is in all three fork dimensions and in a separate fork branch based on `4.21-devel` (`2684956bf`); it is absent from official `main`. The fork also guards trident pair creation when pair production is disabled, fixes a multi-photon-species NBW argument in 2D, restores 2D spin state on restart and warns about explicit charge/mass values replaced by `identify`. These items should be considered independently of the larger spin and pair-boost features.

The pair-boost group in Table 1 spans distinct corrections: Breit–Wheeler boost handling (`3e9f3121f`), muon gamma sampling (`fbb9a9459`), Bethe–Heitler scaling in 3D and then 1D/2D (`92552de69`, `a5716caf2`), and additive optical depths (`99ce3c685`). The table's single row is a tracking convenience, not one patch to cherry-pick.

The custom laser reader has several fork-local corrections: half-cell sampling, MPI-safe file loading, independent storage per laser block and continuous 3D re-slabbing. Official `main` does not contain the custom-file reader, so these are improvements to a fork feature rather than regressions of stock EPOCH. The upstream custom-profile issue #910 was open at the check date; an issue is not an upstream merge.

5 Maintaining this register

For each future entry, record the symptom, affected dimensions, fork commit, official PR or issue (if any), `main` tip checked, an implementation-level test, and whether the change affects numerical results. Refresh the upstream status after each official release or merge; do not infer `main` inclusion from a development-branch PR badge. The command-level method and known limitations are in the accompanying `README.md`.

6 Template for the next entry

Field	Record
Symptom and trigger	What fails, in which dimension and build configuration?
Fork change	Commit, source paths, and whether defaults or numerical results change.
Validation	Smallest reproducer; compiler, ranks, and quantitative acceptance check.
Official status	PR/issue, development branch if applicable, `main` tip and check date.
Remaining limit	Missing tests, uncommitted generators, or dimensional ports.

Immediate refresh targets are the promotion of PR #906 to `main`, a clean-checkout regeneration of the NBW tables, and upstream review of the isolated Higuera–Cary and trident guards. Record a new status date whenever any of these changes.